

CHINESE REMAINDER THEOREM

Theorem 1 (Chinese Remainder Theorem). Suppose we have two integers, m, n such that $\gcd(m, n) = 1$. Then given any $r, s \in \mathbb{Z}$, show that we can find an integer $a \in \mathbb{Z}$ such that:

$$a \equiv r \pmod{m} \text{ and } a \equiv s \pmod{n}.$$

Proof. Given that $\gcd(m, n) = 1$, by what we have shown before, there are integers u, v such that $um + vn = 1$. Now consider $a = sum + rvn$. Note that:

$$a = sum + rvn \equiv rvn = r(1 - um) \equiv r \cdot 1 = r \pmod{m}$$

and similarly $a \equiv s \pmod{n}$ as required. □

Exercise 2. Given the condition of the theorem, that is m, n with $\gcd(m, n) = 1$ and r, s arbitrary, we saw the set

$$S = \{a \in \mathbb{Z} \mid a \equiv r \pmod{m} \text{ and } a \equiv s \pmod{n}\}$$

is non empty. Show that

$$S = \{a_0 + kmn \mid k \in \mathbb{Z}\}$$

for some $a_0 \in \mathbb{Z}$. Equivalently, $a \in S$ if and only if $a \equiv a_0 \pmod{mn}$.

Exercise 3. Consider integers $m_1, m_2, \dots, m_k \in \mathbb{Z}$ such that they are pairwise co-prime, that is, $\gcd(m_i, m_j) = 1$ for all $i \neq j$. For any integers $r_1, r_2, \dots, r_n \in \mathbb{Z}$ show that there is an $a \in \mathbb{Z}$ with $a \equiv r_1 \pmod{m_1}, a \equiv r_2 \pmod{m_2}, \dots, a \equiv r_k \pmod{m_k}$.

Hint: The conditions also imply that $(m_1, m_2 m_3 \dots m_k) = 1$ as well. Now use induction.

Exercise 4. Suppose that $\gcd(m, n) = d$. Assume $r, s \in \mathbb{Z}$ and as before we are interested to find a such that $a \equiv r \pmod{m}$ and $a \equiv s \pmod{n}$. Show that:

- (1) If such an a exist then $r \equiv s \pmod{d}$.
- (2) If $r \equiv s \pmod{d}$, then show that such an a exist.

Hint: Decompose m, n into their prime factors and conclude that we need an a satisfying $a \equiv r \pmod{p^\alpha}$ if $p^\alpha \mid m$ and $a \equiv s \pmod{q^\beta}$ if $q^\beta \mid n$. What about common primes? Now use previous exercise.